Housing Discrimination and Algorithmic Bias: A Data Ethics Project

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Introduction and Historical Context

Housing discrimination in the United States has been a major issue in urban development and property evaluation. Around 1910 and 1955, there was a major problem: racial covenant, a legal issue that did not allow people who were not white to buy properties. This reinforced the racial segregation and limited their opportunities. In 1968, the Fair Housing Act outlawed discriminatory practices, and these patterns can still be seen in today's housing and property values.

The University of Minnesota published the Mapping Prejudice project, where it was documented that over 30,000 racial covenant cases in Hennepin County, showing how modern practices are still shaped by historical practices from back in the day. Areas that are known for taking part in the racial covenants have higher property values, where those who historically did not partake have a deeper wedge between minorities and generational wealth.

Modern mortgage lenders use algorithmic decision-making tools, which usually only provide a yes or no answer. The biggest issue is that the user never understands the reason why each answer is given. These systems are not trained to consider race, but do use property values, income levels, and loan size to determine the race of the individual based on past discrimination. The goal of this project is to explore the correlation between historical discrimination and modern lending patterns through the investigation of data biases and algorithmic biases.

Methodology

We have multiple different methods used in this to analyze and identify how the history of housing discrimination influences modern housing mortgages by using algorithmic systems. While examining counties like Hennepin and Ramsey, using the Mapping Prejudice Project documentation that was implemented between 1910 and 1955.

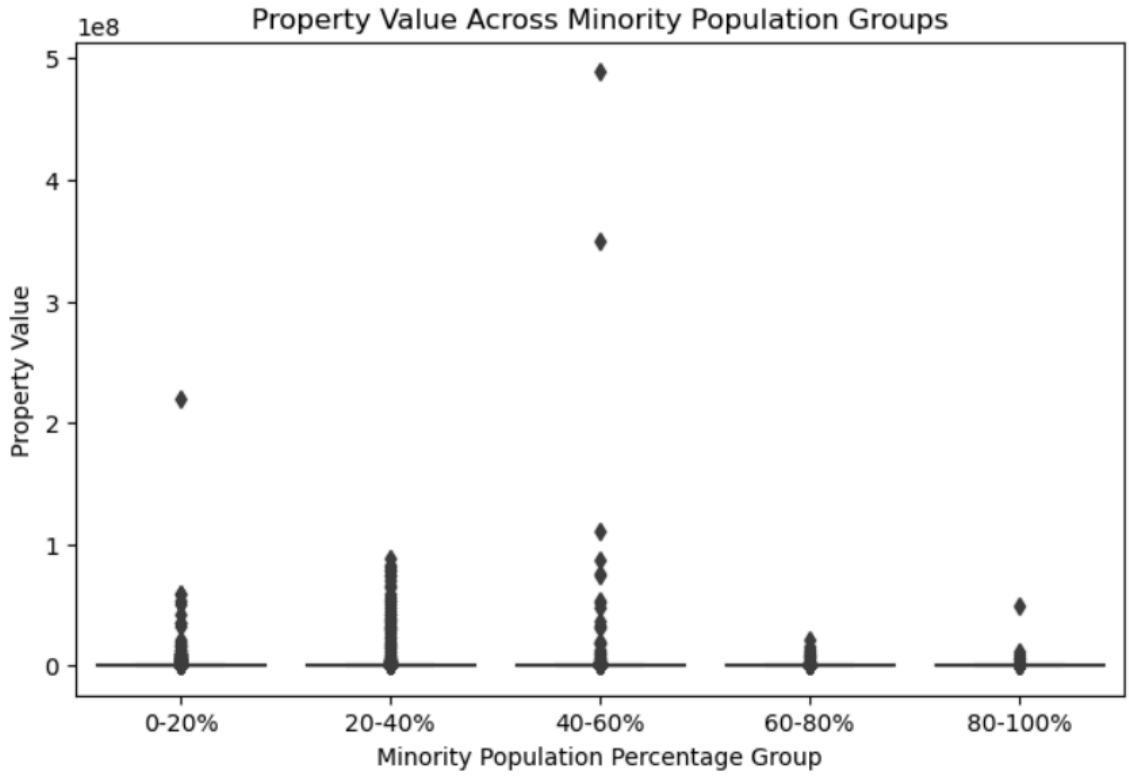
We started by cleaning and normalising the data converting categorical variables to numerical variables. We also derived a few calculations from the present data to produce comparisons like the loan-to-value ratio and debt-to-income ratio, which help understand the relation between covenant density. For analysis, we have utilized a correlation matrix to identify the relation between variables, helping us with density, property, value, loan, and regional demographic properties.

Finally, we have performed a comparative analysis between Hennepin County and Ramsay County with the goal of identifying regional variations in how historical discrimination patterns are present in current mortgage practices. Focusing on approvals, property, evaluation, and loans to find ratios between the group demographics.

Findings

Data bias analysis:

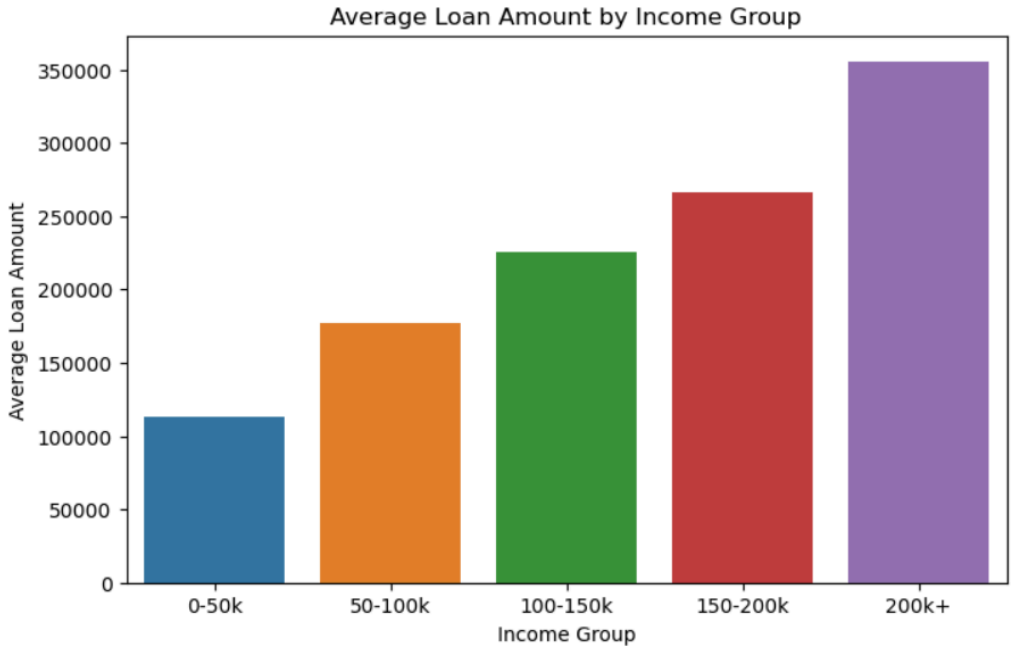
1. Property value distribution:



This graph shows how property values change across neighborhoods with different minority population percentages. It can be seen that areas with higher minority populations tend to have more variation and generally lower property values compared to areas with lower minority populations.

This pattern suggests that housing values are not evenly distributed and may still reflect the effects of historical segregation. Even though race is not directly used, variables like minority population percentage act as indirect indicators, showing how past discrimination can still influence present-day housing data.

2. Income vs Loan amount:

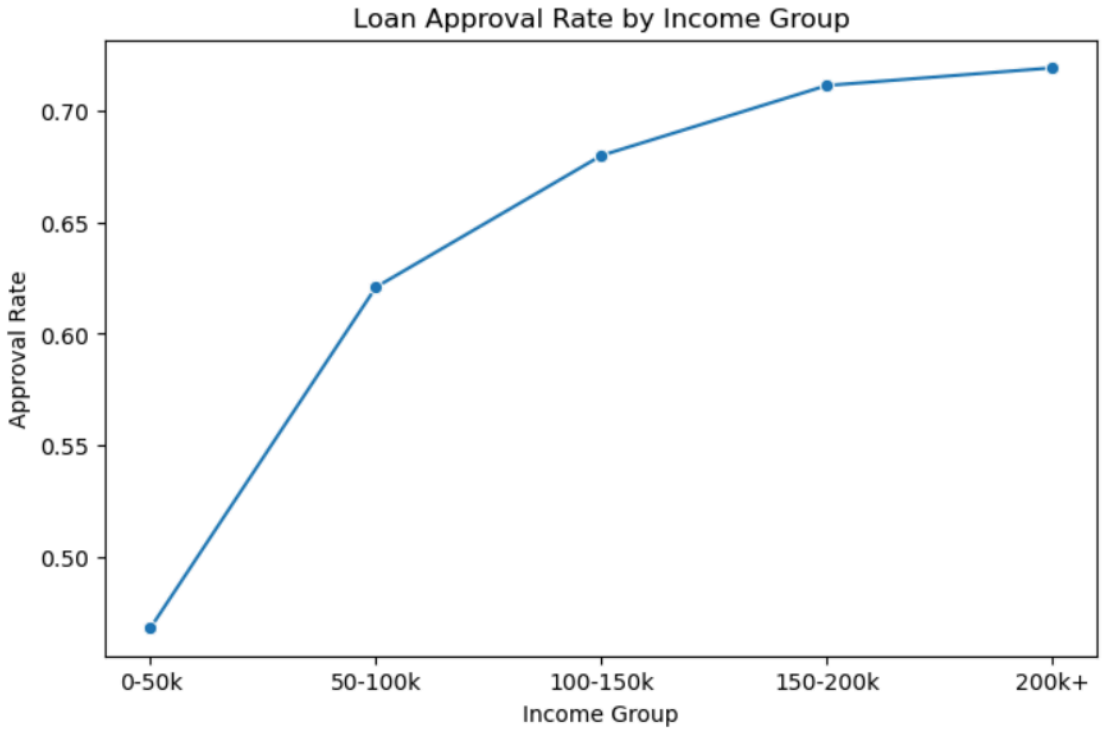


This graph shows the relationship between applicant income and the average loan amount. It can be clearly seen that as income increases, the average loan amount also increases. Applicants in higher income groups receive significantly larger loans compared to those in lower income groups.

This pattern indicates that income plays an important role in determining access to larger mortgage loans. It also highlights how economic differences can influence borrowing capacity, which may contribute to unequal access to housing opportunities.

This also shows that people with higher income have better chances of getting larger loans. Even though income is an important factor, it is often connected to bigger social and economic differences. People with lower income may find it harder to get higher loan amounts, which can limit their housing options. This means that income can indirectly create unequal opportunities in the housing system.

3. Loan approval Rate by Income Group:

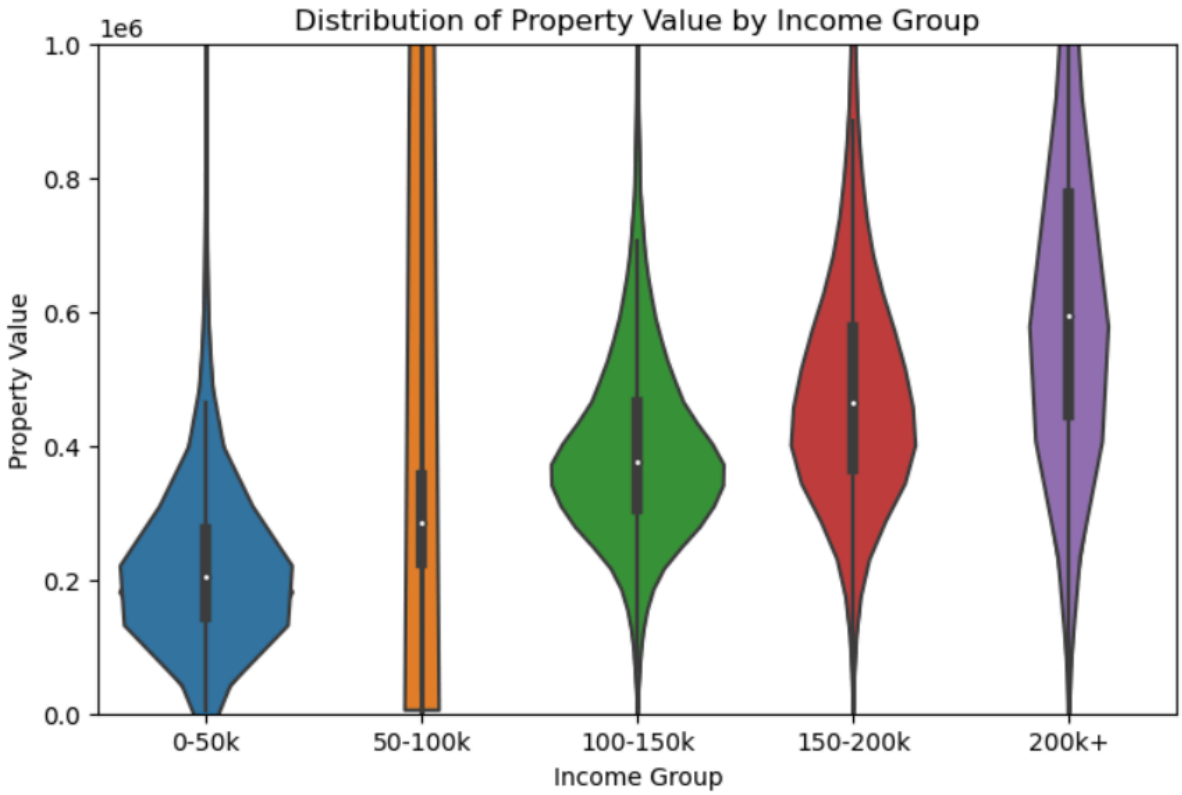


This graph shows how loan approval rates change across different income groups. It can be clearly seen that approval rates increase as income increases. Applicants in lower income groups have lower approval rates, while those in higher income groups are more likely to have their loans approved.

This pattern suggests that income plays a major role in lending decisions. While income is an important financial factor, it can also lead to unequal opportunities, as lower-income applicants may face more difficulty in getting loan approvals.

This also shows how economic differences can influence access to housing. Since income is often linked to broader social inequalities, this pattern may reflect deeper structural differences rather than just individual financial ability.

4. Property value by income group:



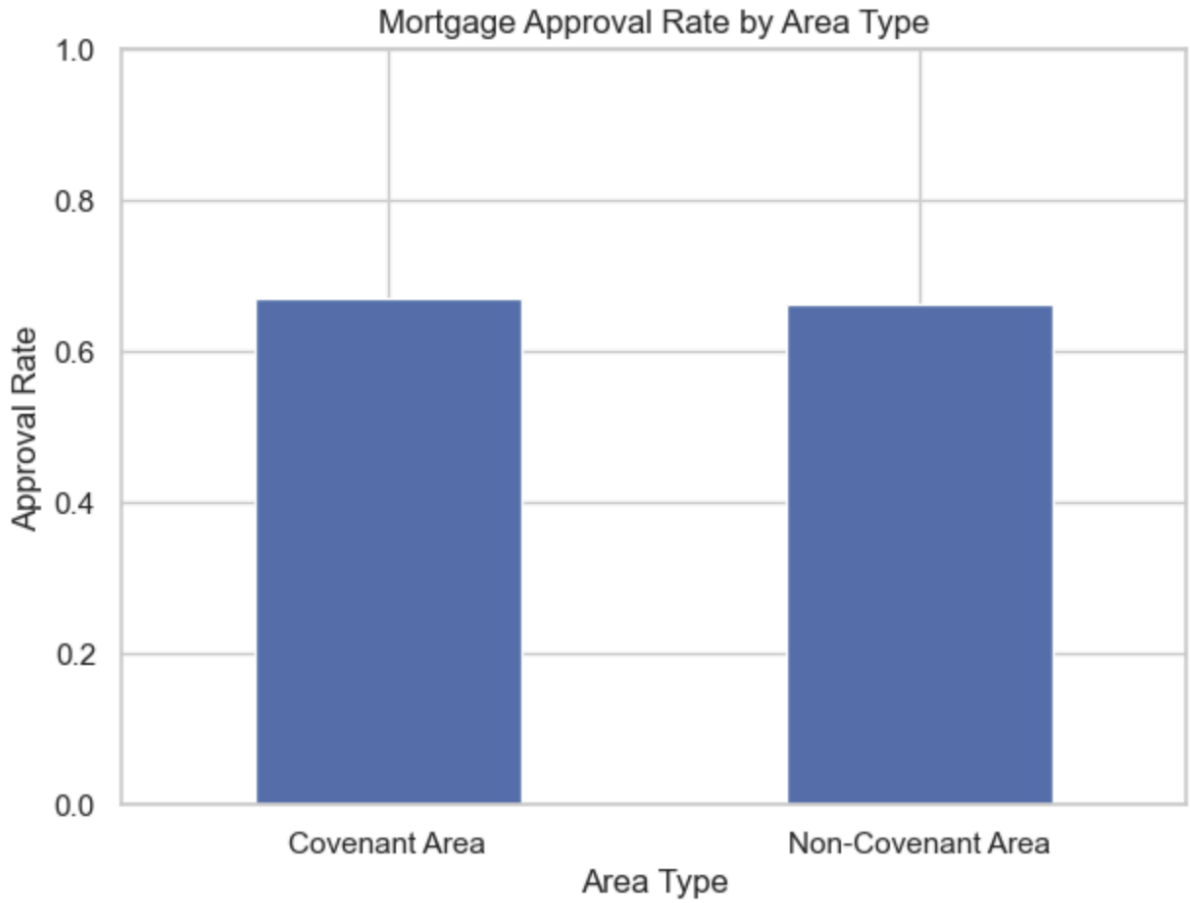
This graph shows how property values are distributed across different income groups. Each shape represents how values are spread within that income category, including the typical range and concentration of values.

It can be observed that as income increases, the distribution shifts toward higher property values. Lower-income groups are mostly concentrated in lower-value properties, while higher-income groups show a wider spread and higher overall values. This indicates that people with higher incomes have access to more expensive housing options.

At the same time, there is some overlap between groups, which suggests that not all individuals follow the same pattern. However, the overall trend clearly shows that income plays a strong role in determining property value.

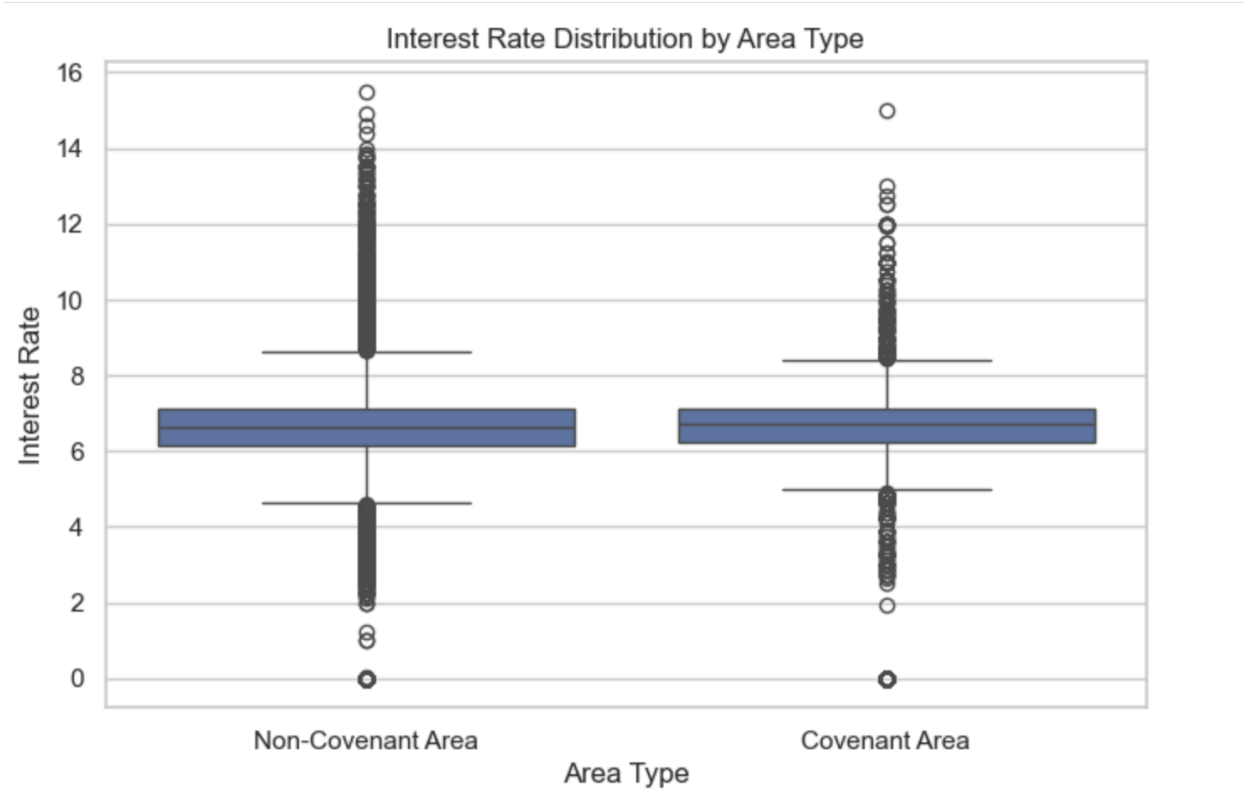
This pattern highlights how economic differences can influence housing opportunities. Since income is often connected to broader social inequalities, this relationship may reflect deeper structural differences in access to housing and wealth.

Algorithmic bias analysis:
1. Mortgage Approval Rate Analysis



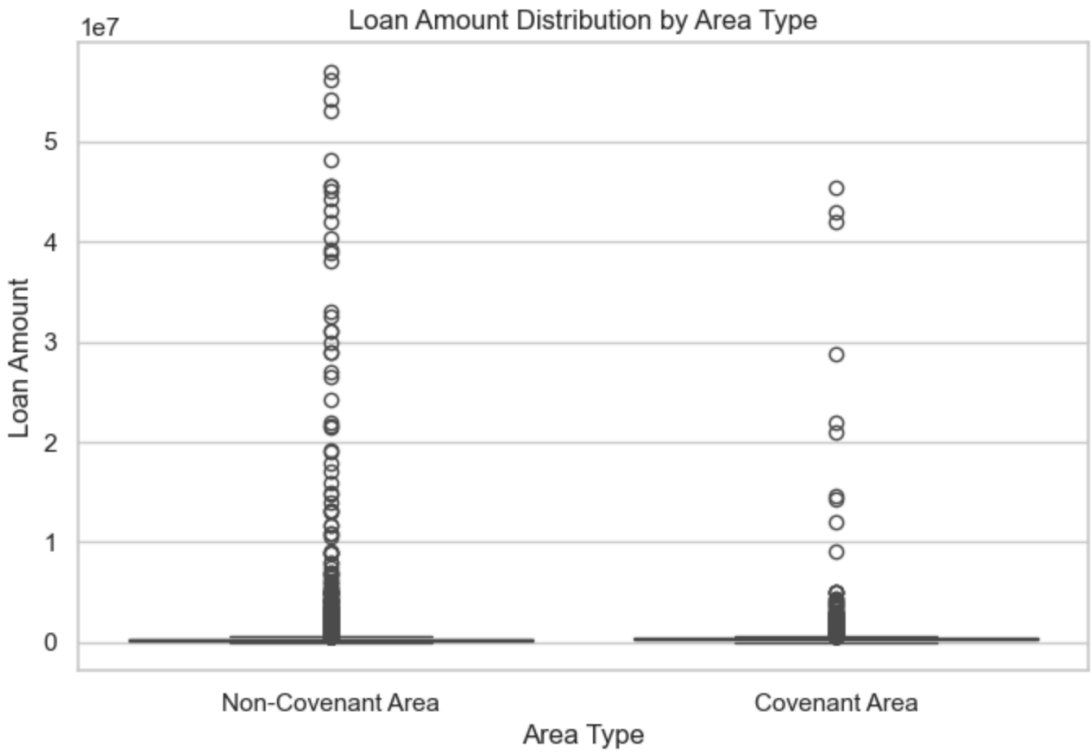
The bar chart compares mortgage approval rates between covenant and non-covenant areas.Both areas show very similar approval rates, around 0.66 to 0.67.There is only a slight difference, with covenant areas having a marginally higher rate.Overall, the visualization suggests that approval rates are nearly equal across the two area types.

2. Interest Rate / Loan Amount Comparison



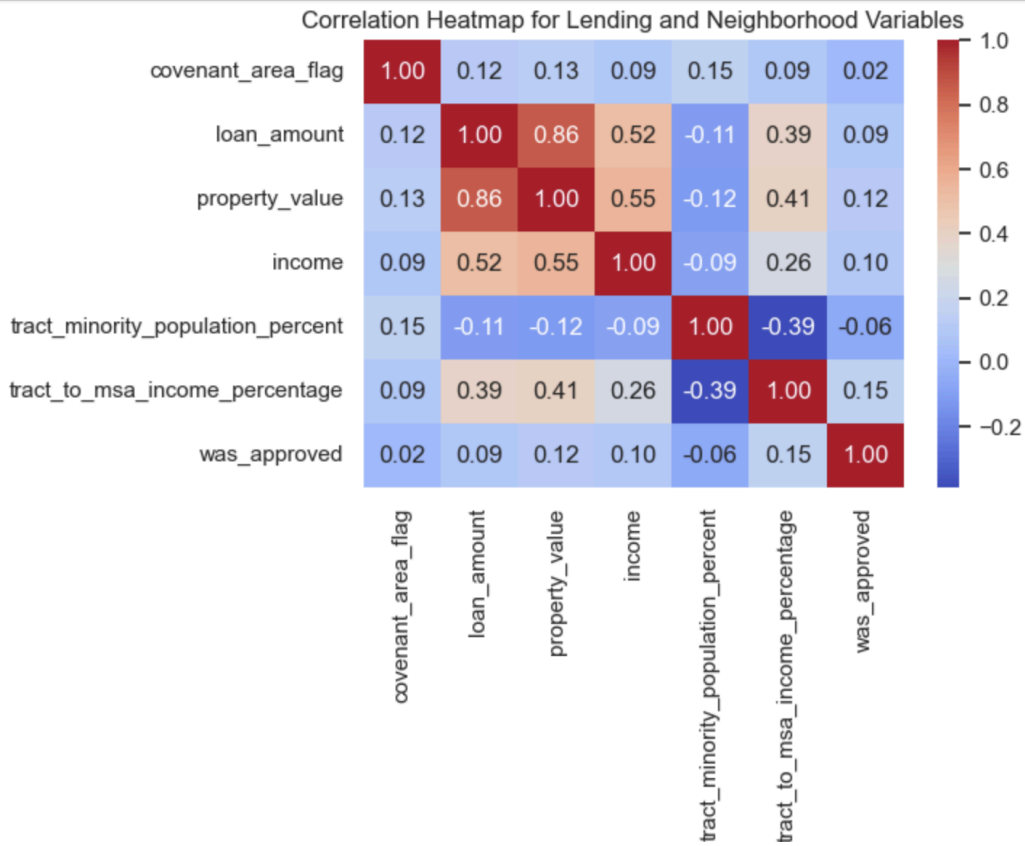
This boxplot compares the spread of interest rates in covenant and non-covenant areas. If covenant areas show higher median interest rates or higher-rate outliers, that may indicate unequal loan conditions. This matters because algorithmic bias is not only about approval or denial, but also about the quality of terms people receive.

3. Loan Amount Comparison



This chart compares loan amounts across the two area types. If applicants in covenant areas receive lower loan amounts on average, it may show that lending systems are still shaped by neighborhood-level inequality. Lower loan amounts can affect home access, refinancing, and wealth-building opportunities.

4. Correlation Analysis for Proxy Variables



The heatmap helps identify which variables are closely related. If covenant area status is associated with property value, income measures, or minority population percentage, this supports the idea that these variables may act as proxies for historical segregation. If approval is also related to those variables, it suggests that algorithms may indirectly reproduce older patterns of discrimination.

Discussion

Implications for modern lending:
Analyzing patterns in historical discrimination in housing from the Ramsay and the Hennepin Counties shows that the liability is present in today's lending system. Even though these systems have been upgraded to have fewer errors, the issue is that the training data has multiple biased values, providing bad information.

For example, machine learning algorithms have shown an increase in showing some risks. Without proper regulation, these models could make the disparities bigger by favoring the privileged groups when making decisions.

Proposed mitigation strategies:

1. **Historical Context needs to be added:** The models should account for the possibility of historical discrimination by having variables that can account for past experiences of injustices.

2. **Incorporating Other Data Sources:** Incorporating different data sources, like rental history, could help provide a more comprehensive understanding of individual applicants with the ability to make payments on a loan rather than classifying them as a minority from a poor area.
3. **Improving Data Collection:** Collaborating between institutions and the government will ensure that the data is more inclusive and representative of all types of people looking for loan approvals.

Limitations and Future Work

Important and impactful biases can be uncovered through the careful exploration of both the past and present data used in developing predictive machine learning algorithms. However, it is not always apparent that this needs to be done. An algorithm can seem fair when tested with a small number of samples, but actually still contains bias, whether through historical or other means. Therefore, it is essential that all data used during algorithm development and training be thoroughly examined, and that all their biases be documented and, if possible, removed. The consequences of failing to do so can have major real-life impacts on people's lives.